Reproducibility Report for Study by Salamons et al. (2026, Animal Behaviour)

Reproduce Figure 2

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Load

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(readxl)
install.packages("patchwork")
```

```
Error in `contrib.url()`:
! trying to use CRAN without setting a mirror
```

```
library(patchwork)
```

```
auditory <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Auditory_Discr
```

```
Rows: 497 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
causal <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Causal_Reasoning
```

```
Rows: 500 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
detour <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Cylinder_Inhibit
```

```
Rows: 498 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
reversal <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Cylinder_Revers
```

```
Rows: 496 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
marker <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Marker_Gesture.cs
```

```
Rows: 503 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
momentary <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Momentary_Poi
```

```
Rows: 495 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
odor <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Odor_Discriminatio
```

```
Rows: 492 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
```

```
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
unsolvable <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Unsolvable_Ey
```

```
Rows: 498 Columns: 12
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (8): Dog ID, Age (wks), Age (days), Total Trials, Mean Look Time, # Tria...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
workingmem <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Working_Memor
```

```
Rows: 493 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
distraction <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Working_Memo
```

```
Rows: 502 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Individual Plots

```
auditory_long <- auditory |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`)
  )

auditory_plot <- auditory_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Auditory Discrimination")
```

```
causal_long <- causal |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`)
  )

causal_plot <- causal_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "Age (weeks)",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
```

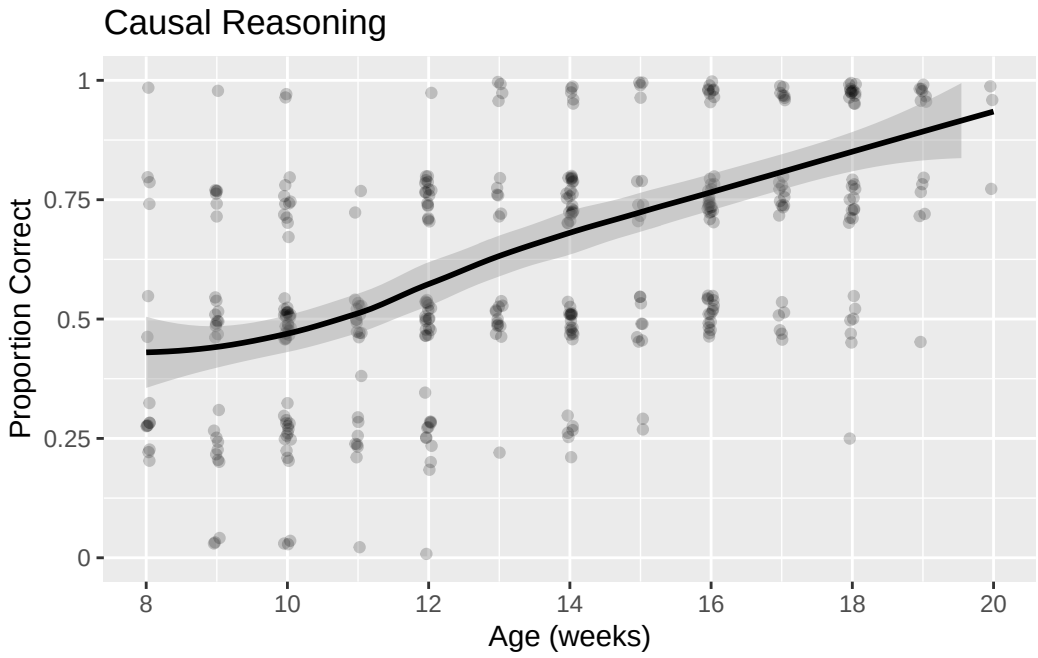
```
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Causal Reasoning")

print(causal_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 103 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
detour_long <- detour |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
```

```
    `Proportion Correct` = as.numeric(`Proportion Correct`)
  )

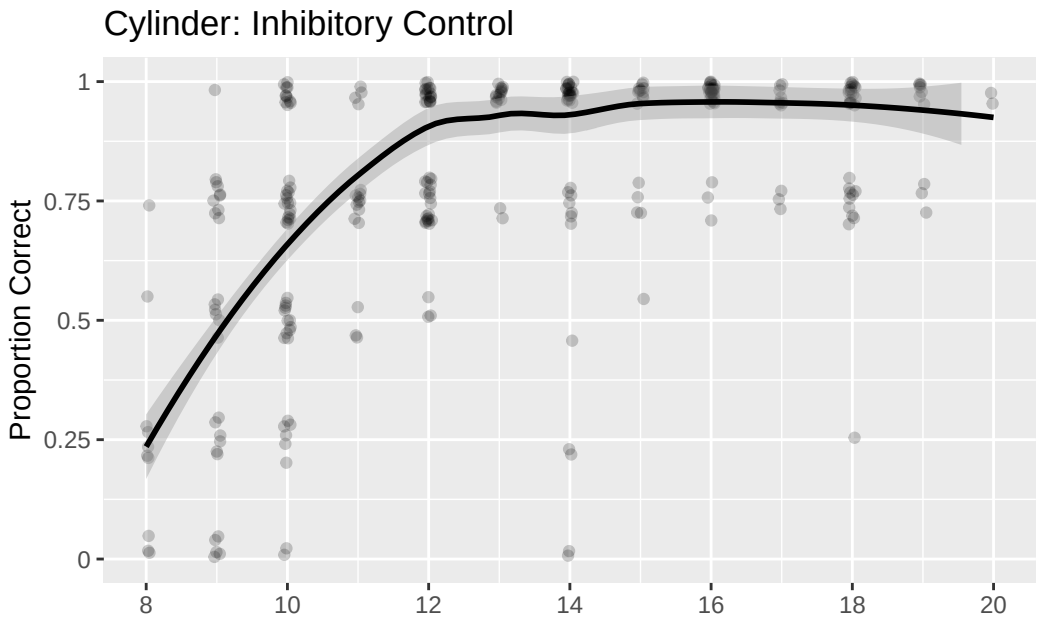
detour_plot <- detour_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Cylinder: Inhibitory Control")

print(detour_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 188 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
distraction_long <- distraction |>
  filter(Status=="Longitudinal") |>
   mutate(
     `Age (wks)` = as.numeric(`Age (wks)`),
     `Proportion Correct` = as.numeric(`Proportion Correct`)
  )

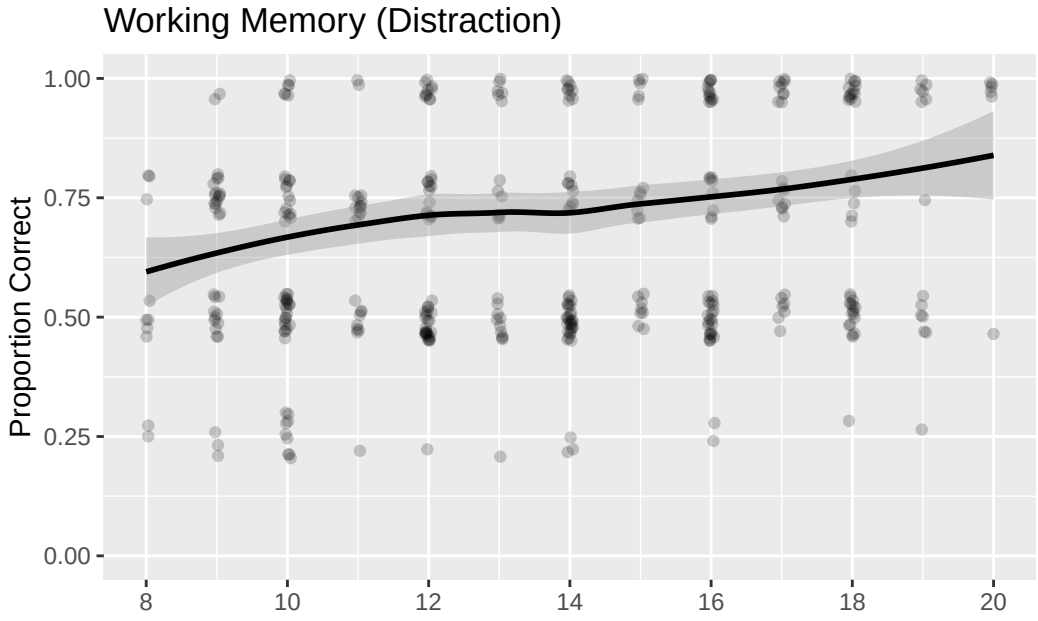
distraction_plot <- distraction_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2),
    labels = as.character
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25)
  ) +
  labs(title = "Working Memory (Distraction)")
```

```
print(distraction_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 102 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
marker_long <- marker |>
  filter(Status=="Longitudinal") |>
   mutate(
     `Age (wks)` = as.numeric(`Age (wks)`),
     `Proportion Correct` = as.numeric(`Proportion Correct`)
  )

marker_plot <- marker_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
```

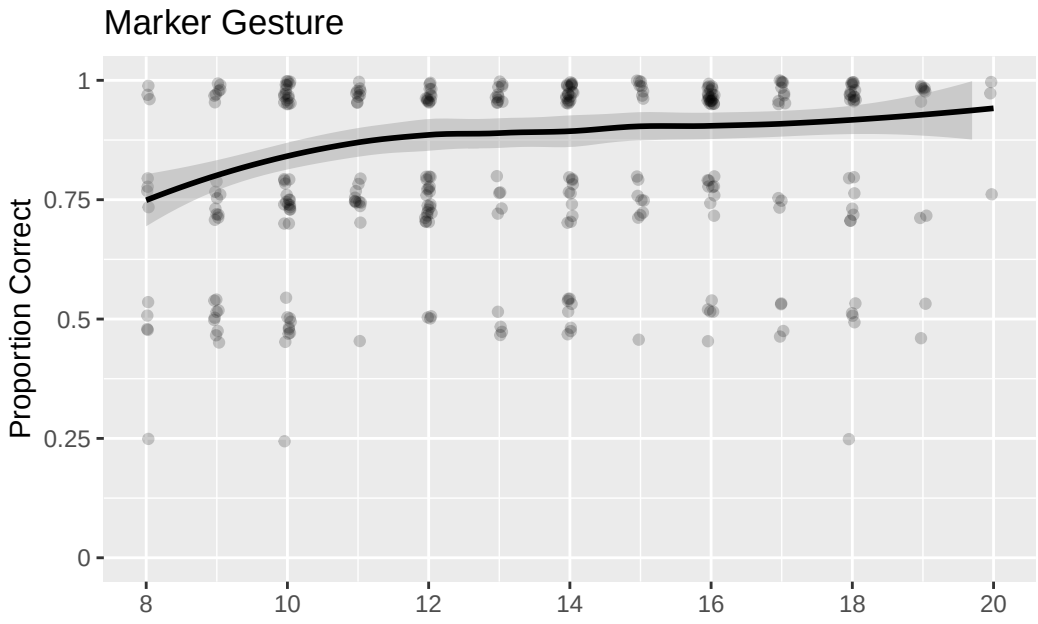
```
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Marker Gesture")

print(marker_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 174 rows containing missing values or values outside the scale range
(`geom_point()`).
```



```r
momentary_long <- momentary |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`)
  )

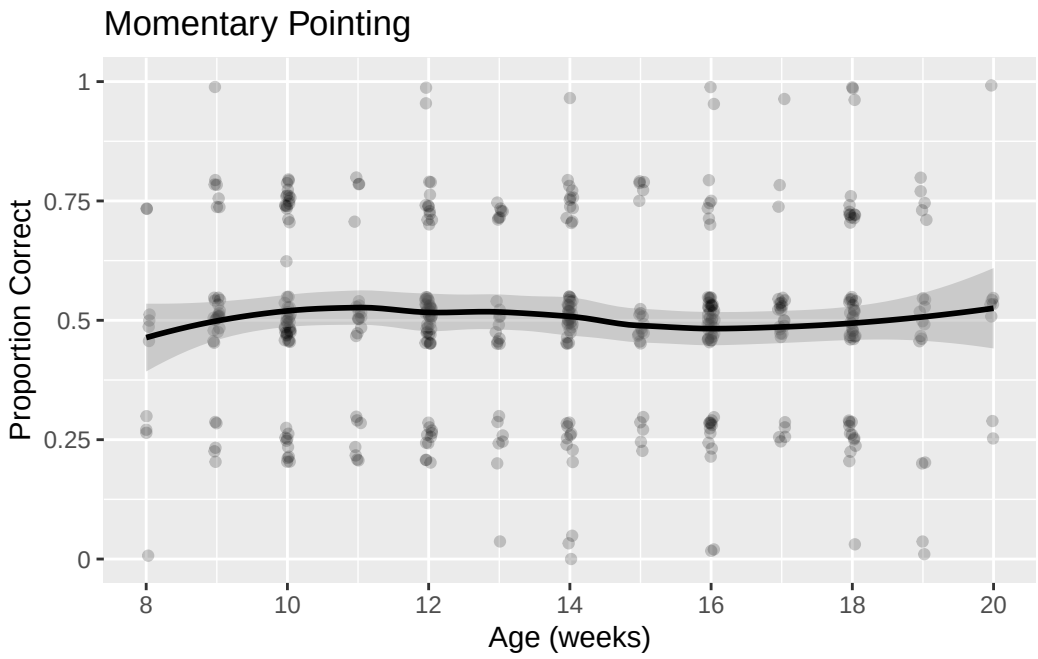
momentary_plot <- momentary_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "Age (weeks)",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Momentary Pointing")

print(momentary_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 35 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
odor_long <- odor |>
  filter(Status=="Longitudinal") |>
   mutate(
     `Age (wks)` = as.numeric(`Age (wks)`),
     `Proportion Correct` = as.numeric(`Proportion Correct`)
  )

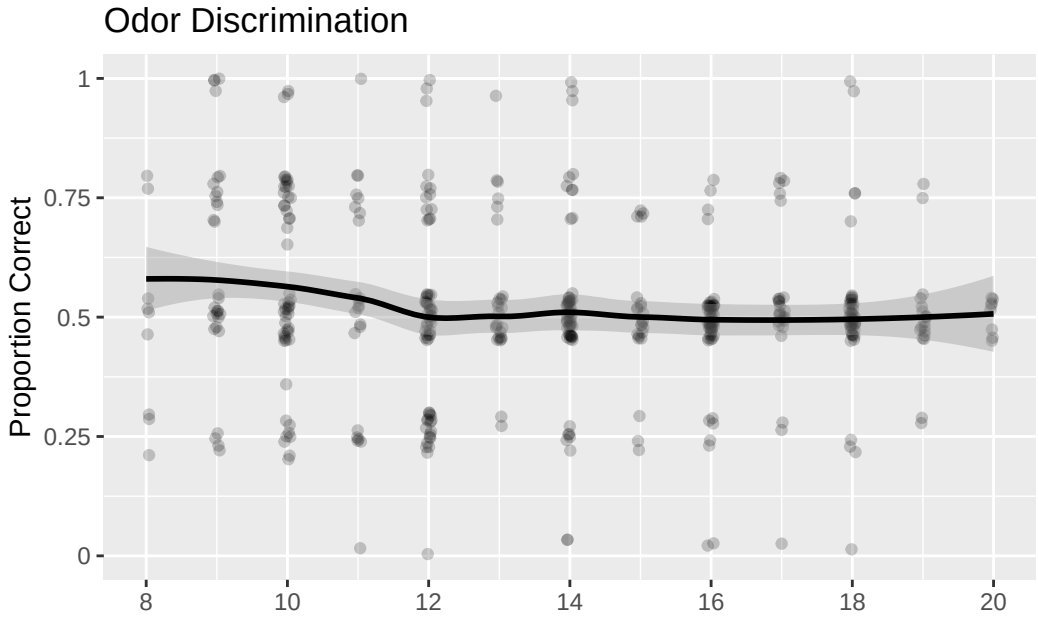
odor_plot <- odor_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Odor Discrimination")
```

```
print(odor_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 38 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
reversal_long <- reversal |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`)
  )

reversal_plot <- reversal_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
```

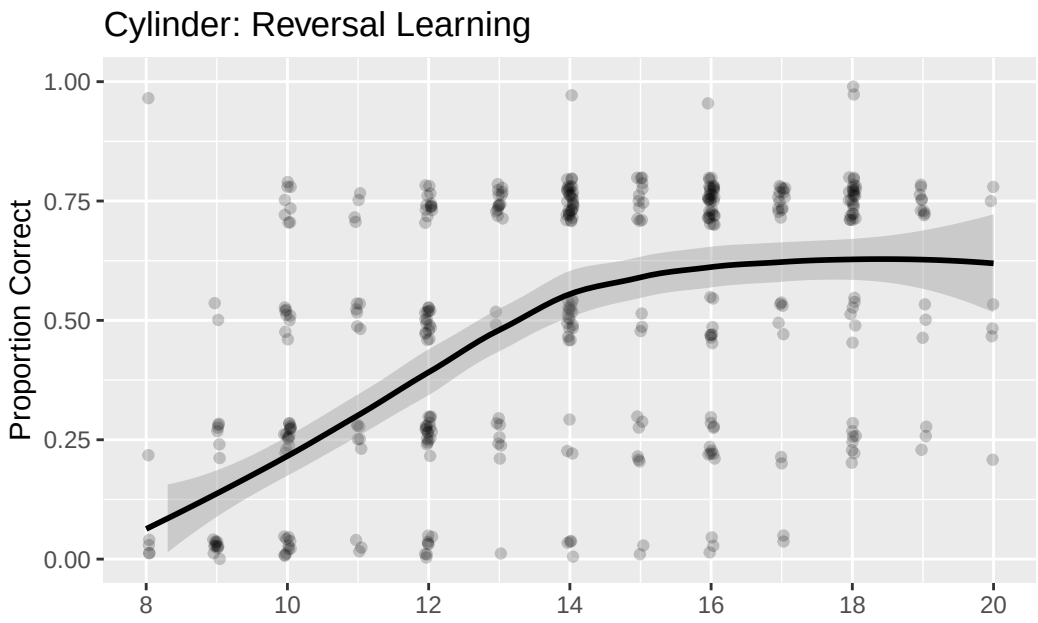
```
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25)
  ) +
  labs(title = "Cylinder: Reversal Learning")

print(reversal_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 87 rows containing missing values or values outside the scale range
(`geom_point()`).
```



```r
unsolvable_long <- unsolvable |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Trials Looked` = as.numeric(`Proportion Trials Looked`),
    `Mean Look Time` = as.numeric(`Mean Look Time`)
  )

proplooked_plot <- unsolvable_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Trials Looked`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "Age (weeks)",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Unsolvable (Proportion Looked)")

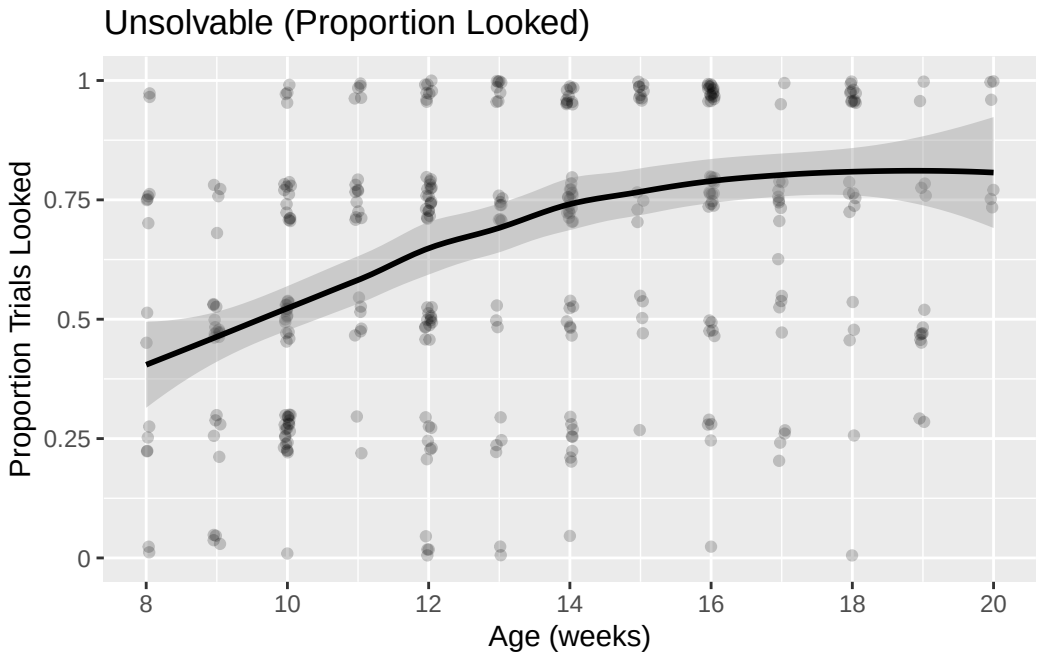
looktime_plot <- unsolvable_long|>
  ggplot(aes(x=`Age (wks)`, y = `Mean Look Time`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    name = "Mean Look Time (s)",
    limits = c(0, 20),
    breaks = seq(0, 20, by = 5)
  ) +
  labs(title = "Unsolvable (Looking Time)")

print(proplooked_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 121 rows containing missing values or values outside the scale range
(`geom_point()`).
```

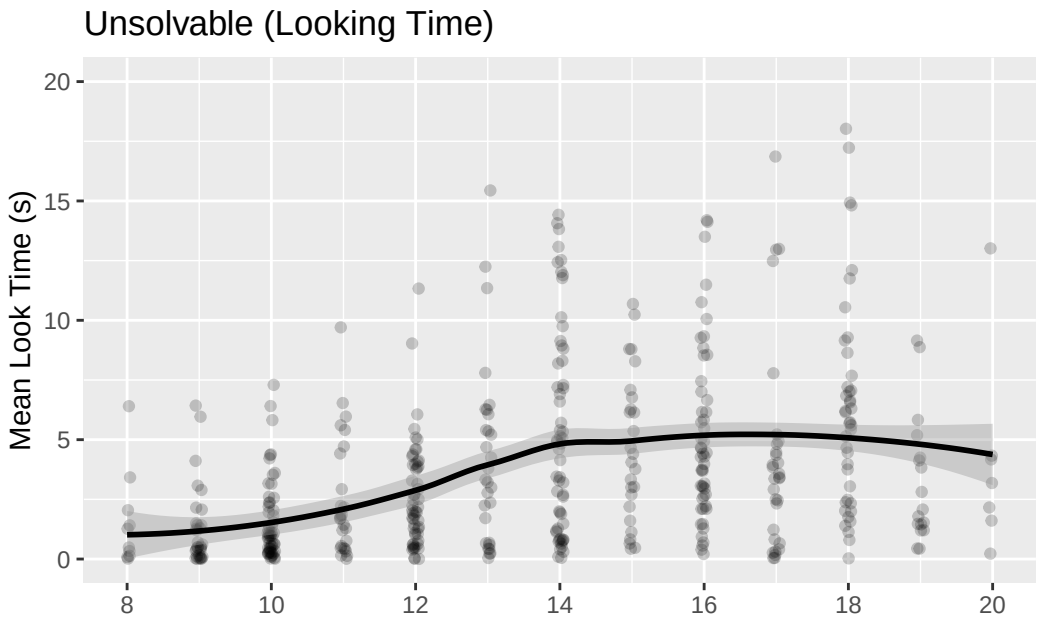



```
print(looktime_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 3 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 41 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
workingmem_long <- workingmem |>
  filter(Status=="Longitudinal") |>
   mutate(
     `Age (wks)` = as.numeric(`Age (wks)`),
     `Proportion Correct` = as.numeric(`Proportion Correct`)
  )

workingmem_plot <- workingmem_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name ="",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Working Memory (Delay)")
```

```
print(workingmem_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 145 rows containing missing values or values outside the scale range
(`geom_point()`).
```

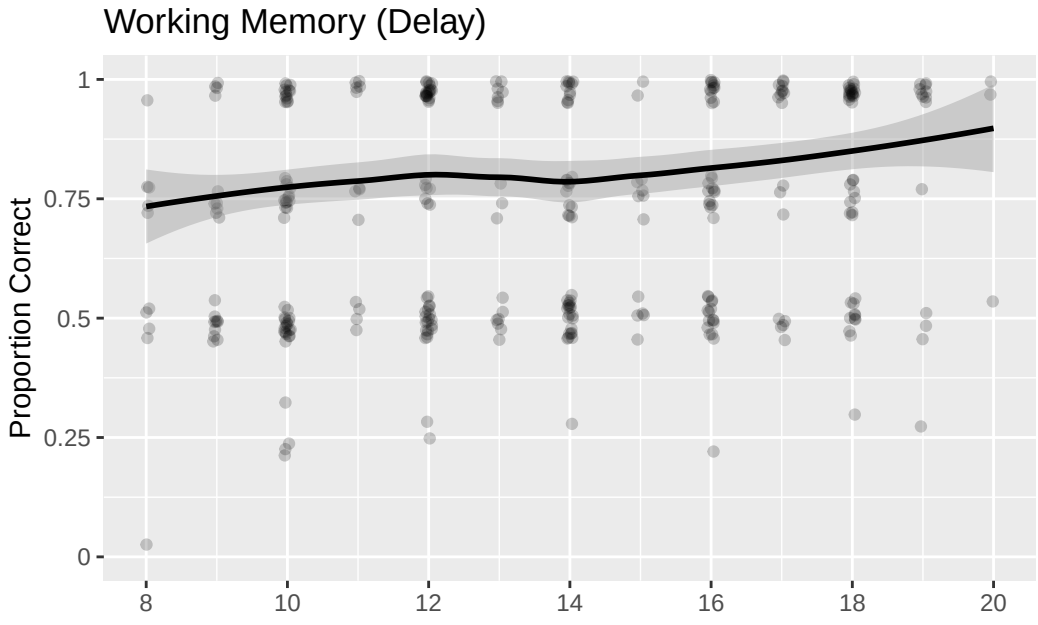


Figure 2

```
(marker_plot   | workingmem_plot | distraction_plot) /
(auditory_plot | odor_plot       | detour_plot)      /
(reversal_plot | looktime_plot   | proplooked_plot)  /
(causal_plot   | momentary_plot  | plot_spacer())&
  theme_grey(base_family = "serif") +
  theme(
```

```
    plot.title = element_text(hjust = 0.5, size = 11),
    axis.line  = element_line(colour = "black"),
    axis.title = element_text(size = 10),
    axis.text  = element_text(size = 9, colour = "black")
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 173 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 143 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 98 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 68 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 42 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 190 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 92 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 3 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 33 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 121 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'

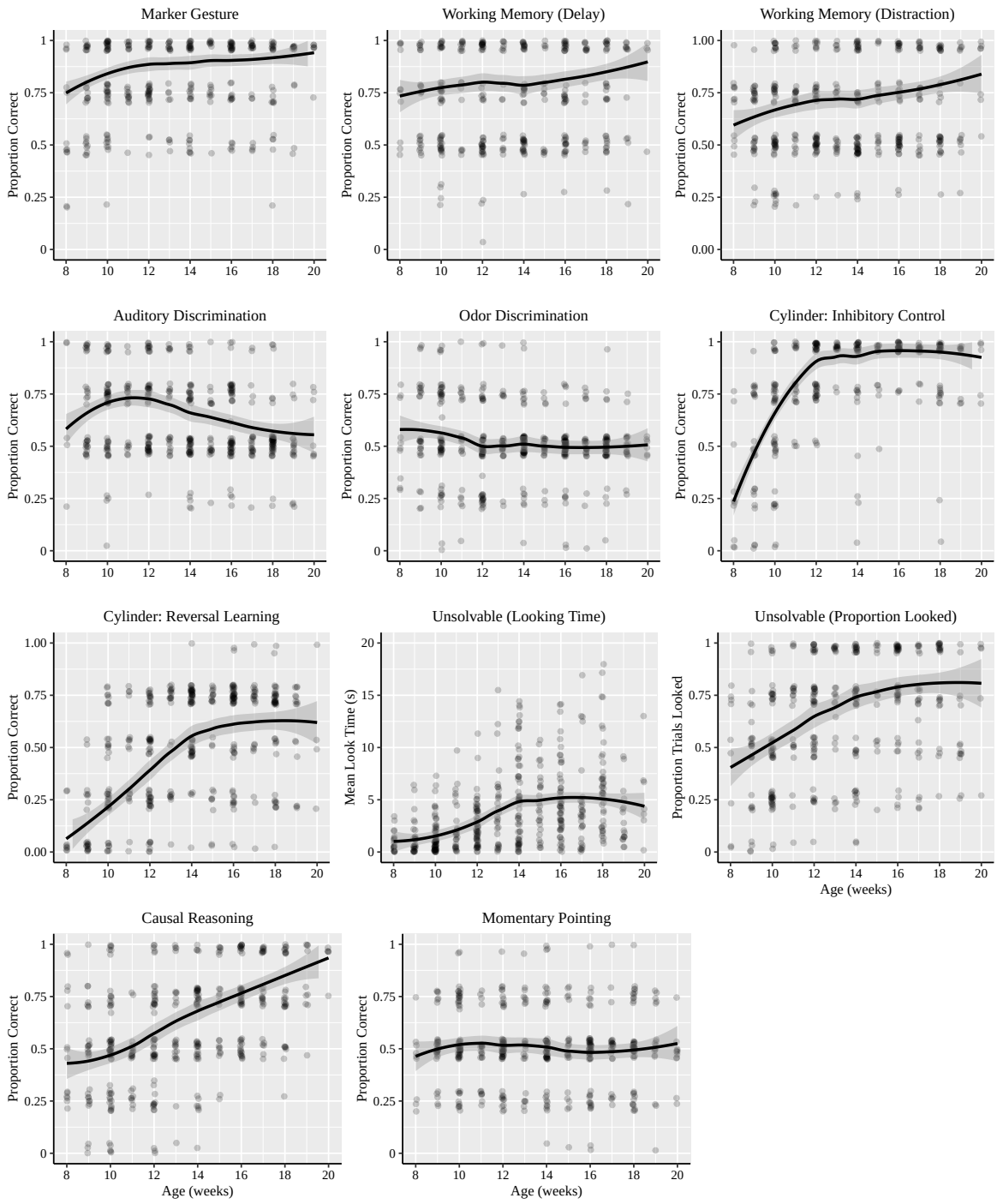
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 90 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 37 rows containing missing values or values outside the scale range
(`geom_point()`).
```


Marker Gesture
Working Memory (Delay)
Working Memory (Distraction)
Auditory Discrimination
Odor Discrimination
Cylinder: Inhibitory Control
Cylinder: Reversal Learning
Unsolvable (Looking Time)
Unsolvable (Proportion Looked)
Causal Reasoning
Momentary Pointing
Proportion Correct
Mean Look Time (s)
Proportion Trials Looked
Age (weeks)